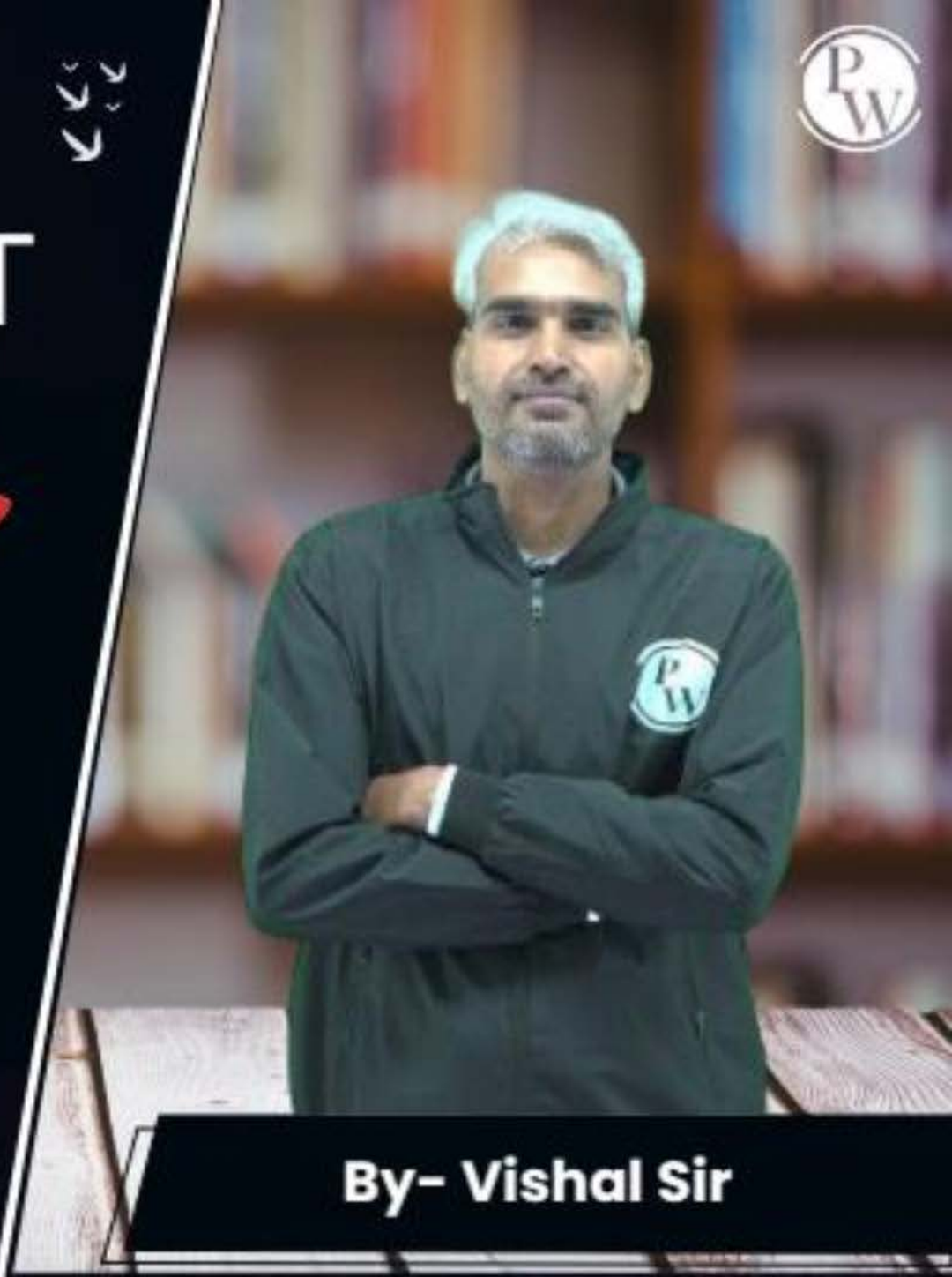
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



- Topic: Multi-set
- Topic: Cartesian product
- Topic: Relation and types of relations

Slide

Topics to be Covered



- Topic — Reflexive relation
- Topic — Irreflexive relation
- Topic — Symmetric relation
- Topic — Anti-symmetric relation
- Topic — Asymmetric relation
- Topic — Transitive relation

Slide

Cartesian Product / Cross Product

Let A & B are two sets.
Cartesian product of A and B is denoted by 'A x B' and it is defined as,

$$A \times B = \{ (a, b) \mid a \in A \text{ and } b \in B \}$$

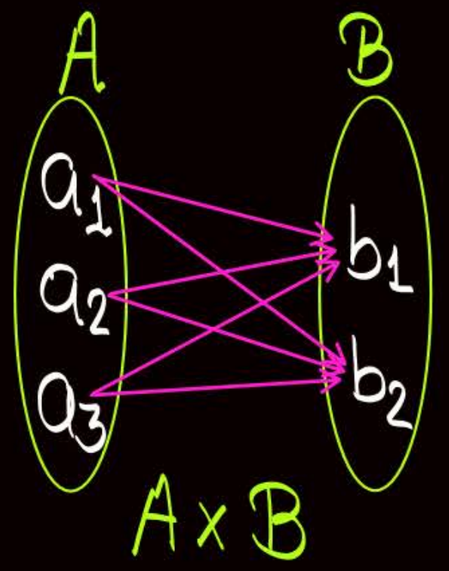
Order pair
i.e., order is important
$(a, b) \neq (b, a)$

first element will be from 1st set

2nd element of order pair will be from 2nd set.

eg: Let $A = \{a_1, a_2, a_3\}$

$B = \{b_1, b_2\}$



$A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2)\}$

$B \times A = \{(b_1, a_1), (b_1, a_2), (b_1, a_3), (b_2, a_1), (b_2, a_2), (b_2, a_3)\}$

We know, $(a_1, b_1) \neq (b_1, a_1)$ $\therefore$ in general $A \times B \neq B \times A$

In general,

$$A \times B \neq B \times A$$

- "$A \times B = B \times A$" if and only if

  (i) Either $A = B$

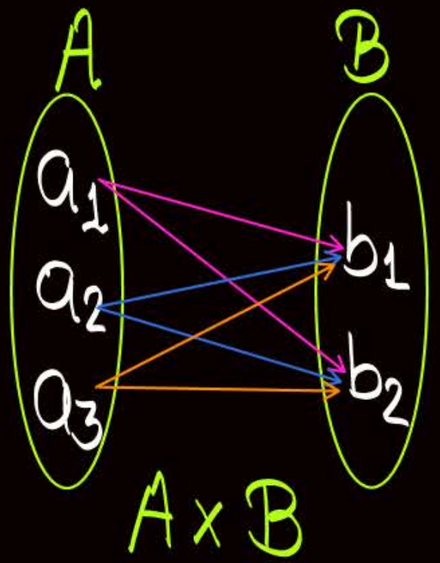
  Or (ii) At least one set out of A or B is an Empty set.

If at least one of A or B is an Empty set,
then $A \times B = B \times A : \{\ \}$

Result will be an Empty set

eg: Let $A = \{a_1, a_2, a_3\}$

$B = \{b_1, b_2\}$



$A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2)\}$

Note:- We can observe that in $A \times B$ every element of set A relates with every element of set B.

∴ If $|A| = m$ & $|B| = n$

then $|A \times B| = m.n. = |A| \cdot |B|$



Topic : Relation

- A relation from set A to set B defines that how exactly the elements of set A relates with the elements of set B.

* In Cartesian product 'A × B' Every element of set A relates with every element of set B.
  i.e 'A × B' is the universal set w.r.t. relations from set A to set B.

> Any relation from set A to set B is a subset of A × B

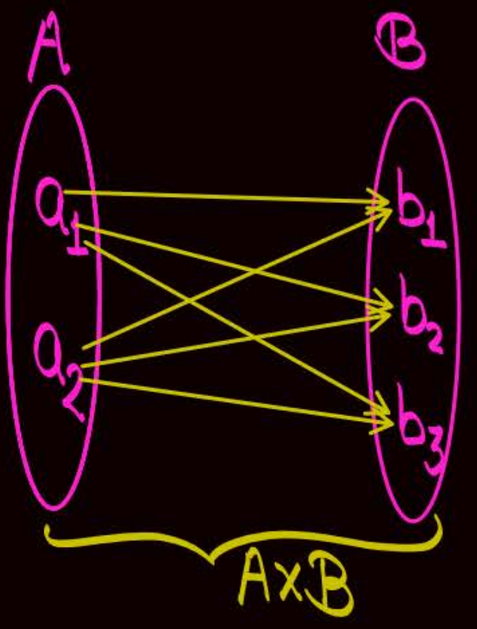
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eg:-

$A = \{a_1, a_2\}$

$B = \{b_1, b_2, b_3\}$

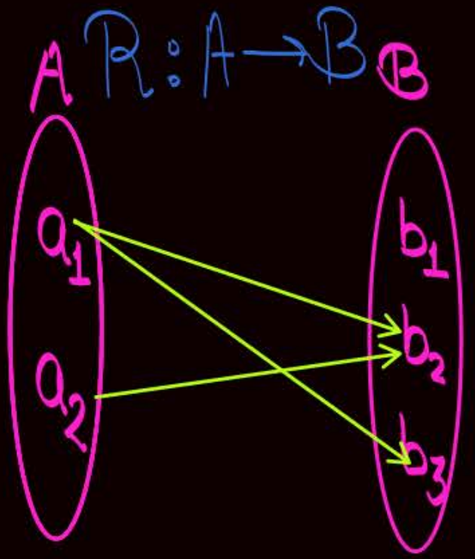
* $A \times B = \{(a_1, b_1), (a_1, b_2), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_2, b_3)\}$



+ Let 'R' denotes a relation from set A to set B defined as

$R = \{(a_1, b_2), (a_1, b_3), (a_2, b_2)\}$

We can observe that all order pairs of relation R are member of $A \times B$, ie R is a subset of $A \times B$



* Number of different relations possible from set A to set B

- Let A and B are two sets such that $|A| = m$ and $|B| = n$, then

* How many different relations are possible from set A to set B?

- Every relation from set A to set B is a subset of $A \times B$

∴ Total no. of relations possible from set A to set B = Number of subsets of $A \times B$

$$= 2^{|A \times B|} = 2^{|A| \cdot |B|}$$

$$= 2^{m \cdot n}$$

- One subset of 'A x B' is Empty set.
- This Empty subset of A x B is also a relation from set A to set B; and it is called "Empty relation" from set A to set B.



Topic : Relation

☆ A relation from set A to set A it self is called a relation on set A

* Every relation on set A is a subset of "A × A"

- If $|A| = n$, then

Number of relations possible on set A = Number of subsets of A×A

$= 2^{|A \times A|} = 2^{|A| \cdot |A|}$

$= 2^{n \cdot n} = 2^{(n^2)}$

Slide



Topic : Types of Relations

All these relations are defined on same set {i.e from set A to set A itself}

(1) Diagonal Relation (Identity Relation)

(2) Reflexive Relation

(3) Irreflexive Relation

(4) Symmetric Relation

(5) Anti-symmetric Relation

(6) Asymmetric Relation

(7) Transitive Relation

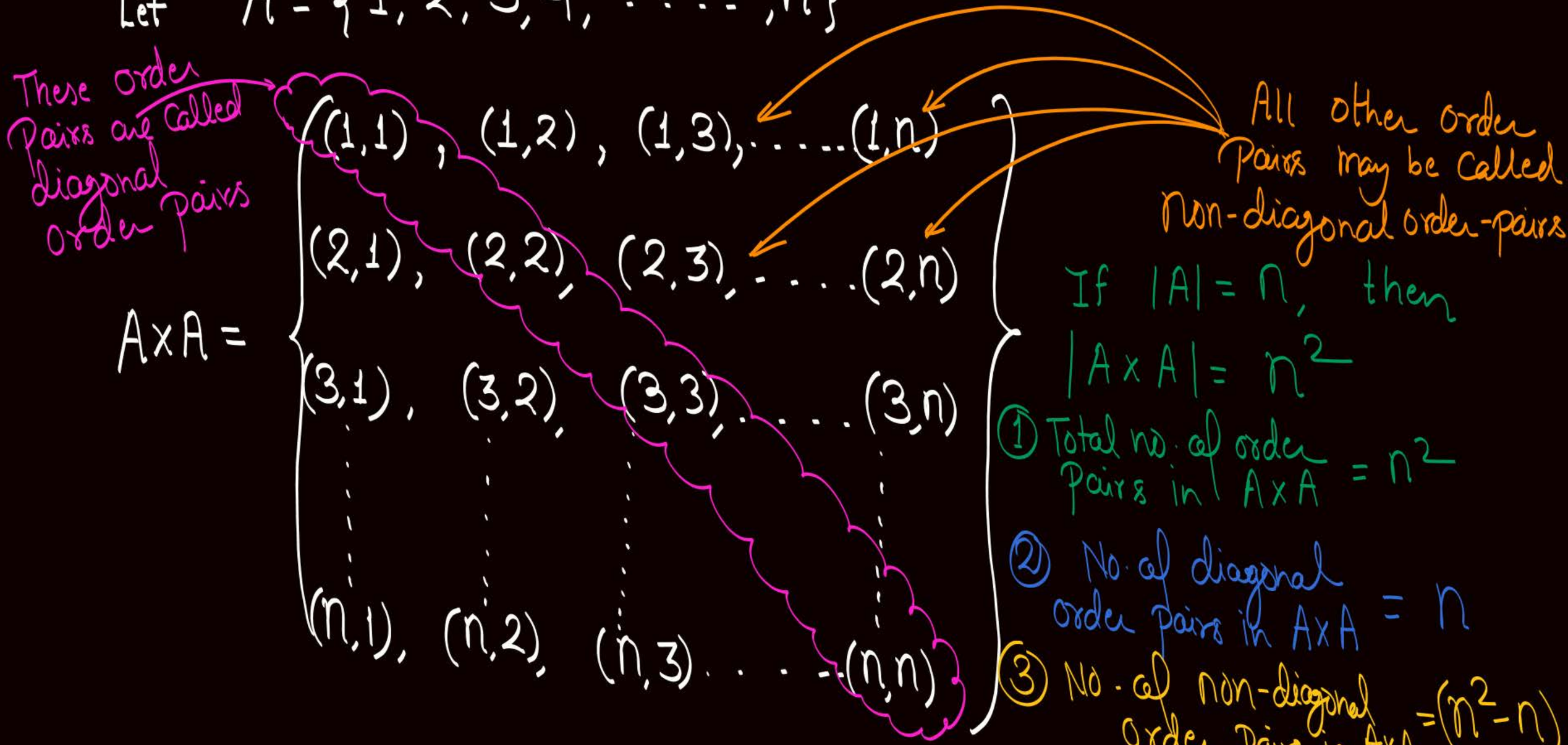
(8) Complement of a relation

(9) Inverse of a relation

(10) Composite of two relations

Slide

Let $A = \{1, 2, 3, 4, \ldots, n\}$



If $|A| = n$, then

$|A \times A| = n^2$

① Total no. of order pairs in $A \times A = n^2$

② No. of diagonal order pairs in $A \times A = n$

③ No. of non-diagonal order pairs in $A \times A = (n^2 - n)$



Topic : Diagonal Relation (Identity Relation)

- A diagonal relation on set A (i.e. identity relation on set A) can be denoted by Δ_A (or I_A) and it is defined as,

$$\Delta_A = \{(a,a) \mid a \in A\}$$ ← it is the definition of the set.

eg. Let $A = \{1, 2, 3\}$

then $\Delta_A = \{(1,1), (2,2), (3,3)\}$ it is diagonal relation on set A.

$R_1 = \{(1,1), (2,2)\}$ $3 \in A$, but $(3,3) \notin R_1$ ∴ R_1 is not a diagonal relation

$R_2 = \{(1,1), (2,2), (3,3), (2,3)\}$, here $(2,3) \in R_2$ but $2 \neq 3$ ∴ Not a diagonal relation

Slide

Note:- On any given set A, Exactly "one" diagonal relation is possible

Note: Cardinality of diagonal relation on set A = $|A|$

Let $A = \{1, 2, 3, 4, \ldots, n\}$

$$A \times A = \begin{Bmatrix} (1,1), (1,2), (1,3), \ldots (1,n) \\ (2,1), (2,2), (2,3), \ldots (2,n) \\ (3,1), (3,2), (3,3), \ldots (3,n) \\ \vdots \quad \vdots \quad \vdots \quad \vdots \\ (n,1), (n,2), (n,3) \ldots (n,n) \end{Bmatrix}$$

Today's Topic



Topic : Reflexive Relation

A relation R on set 'A' is called a reflexive relation if and only if,

$$\begin{cases} a\,R\,a, \ \forall a \in A \\ \text{i.e., } (a,a) \in R, \ \forall a \in A \end{cases}$$

It is a Constraint on the relation for relation to be called a reflexive relation

$a\,R\,b$ is read as 'a relates to b' w.r.t. rel^n R
i.e. $(a,b) \in R$

$a\,\not{R}\,b$ is read as 'a does not relate with b'
i.e. $(a,b) \notin R$

Slide

Eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{(1,1), (2,2), (3,3)\}$

it is a diagonal relation on set A

R_1 is diagonal relation as well as a reflexive relation on set A.

② $R_2 = \{(1,1), (2,2), (3,1), (3,3)\}$ R_2 is a reflexive relation on set A, but not a diagonal relation

③ $R_3 = \{(1,1), (2,2)\}$ $3 \in A$, but $(3,3) \notin R_3$ $\therefore$ R_3 is not reflexive

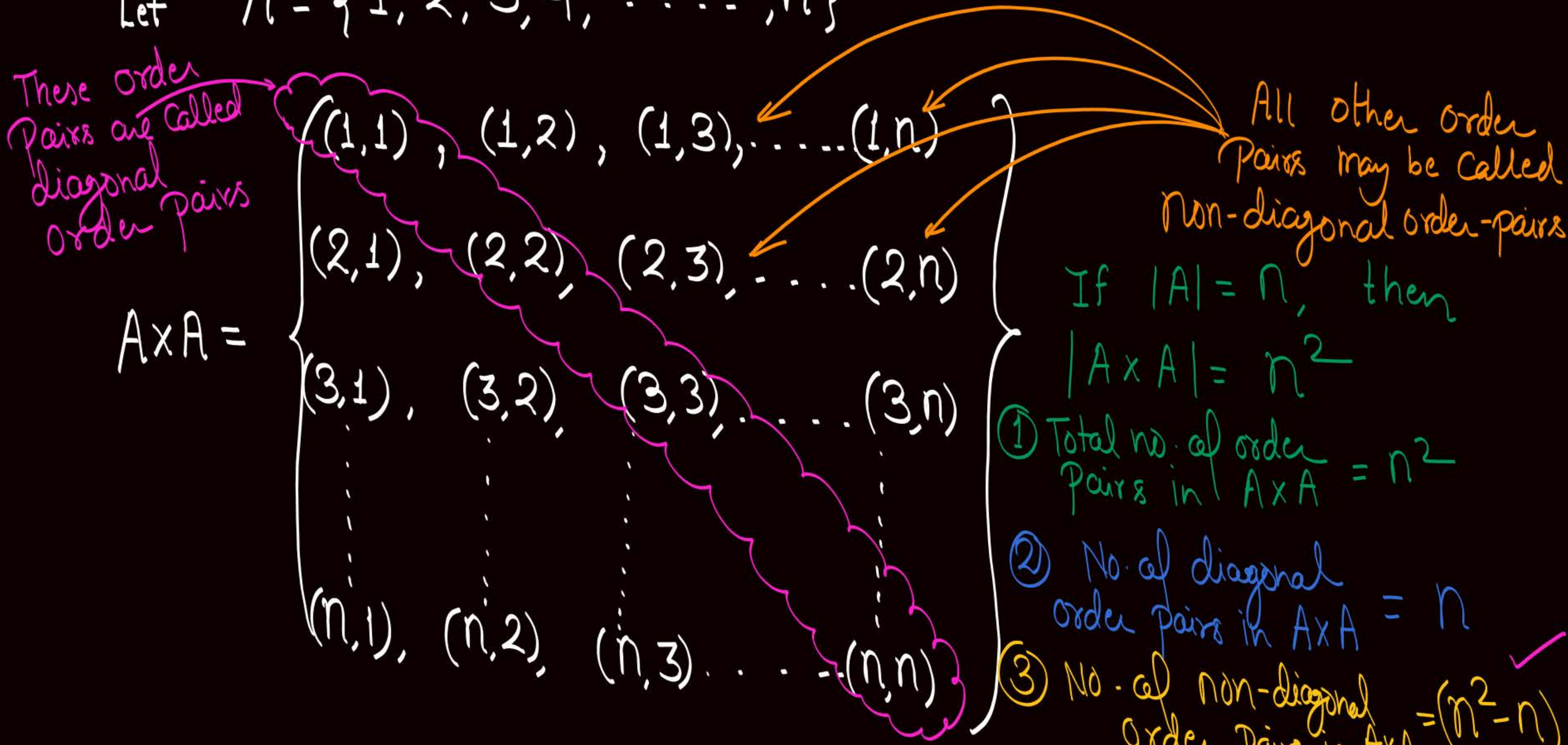
④ $R_4 = \{(1,2), (2,3), (3,1)\}$ R_4 is not reflexive

Note:- ① Every diagonal relation on set A is also a reflexive relation on set A, but every reflexive relation on set A need not be a diagonal relation on set A.

② Diagonal relation on set A is the smallest reflexive relation on set A.

③ Largest reflexive relation on set A is "A×A"

Let $A = \{1, 2, 3, 4, \ldots, n\}$



If $|A| = n$, then

$|A \times A| = n^2$

① Total no. of order pairs in $A \times A = n^2$

② No. of diagonal order pairs in $A \times A = n$

③ No. of non-diagonal order pairs in $A \times A = (n^2 - n)$

Q: Let $A = \{1, 2, 3, \ldots, n\}$
How many reflexive relation are possible on set A where $|A| = n$.

Reflexive Relation = (Choose all 'n' out of 'n' diagonal order pairs) and (Along with all diagonal order pairs, any no. of non-diagonal order pairs can be chosen)

$$\#\text{Reflexive Relation} = {}^{n}C_{n} * \left({}^{n^2-n}C_0 + {}^{n^2-n}C_1 + \cdots + {}^{n^2-n}C_{n^2-n}\right)$$

$$= 1 * 2^{n^2-n}$$

$$= \boxed{2^{n^2-n}}$$

Another approach to understand no. of reflexive rel^n on set A.

Let $A = \{1, 2, 3, 4, \ldots, n\}$

$$A \times A = \begin{cases} (1,1), (1,2), (1,3), \ldots (1,n) \\ (2,1), (2,2), (2,3), \ldots (2,n) \\ (3,1), (3,2), (3,3), \ldots (3,n) \\ \vdots \\ (n,1), (n,2), (n,3), \ldots (n,n) \end{cases}$$

1 * 2 * 2 * ... * 1

2 * 2 * 2 * ... * 2

2 * 2 * 2 * ... * 1

Reflexive Rel^n

$= \underbrace{1 * 1 * \cdots * 1}_{n \text{ times } 1} * \underbrace{2 * 2 * 2 * \cdots * 2}_{(n^2 - n) \text{ times } 2}$

$= 2^{n^2 - n}$



Topic : Irreflexive Relation

A relation R on set A is called irreflexive relation if and only if

$a \not R a, \quad \forall a \in A$

i.e. $(a, a) \notin R, \quad \forall a \in A$

} i.e. none of the diagonal order pair should be present in the relation

Slide

eg: Let, $A = \{1, 2, 3\}$

① $R_1 = \{\}$ — Empty relation

Empty relation is the smallest irreflexive relation on any set A.

② $R_2 = \{(1,2), (2,3), (3,1)\}$ R_2 is an irreflexive relation

③ $R_3 = \{(1,2), (2,1), (2,2), (3,2)\}$

$(2,2) \in R_3 \therefore R_3$ is not an irreflexive relⁿ

$(1,1) \notin R_3 \therefore R_3$ is not reflexive as well

Let 'A' is any finite set,

then "$A \times A - \Delta_A$" is the largest irreflexive relation set A

Q: Let $A = \{1, 2, 3, \ldots, n\}$

How many "irreflexive relation" are possible on set A where $|A| = n$.

Irreflexive relation = (Choose '0' out of 'n' diagonal order pairs) and (any no. of non-diagonal order pairs may be present)

$$\#\text{ Irreflexive relation} = {}^{n}C_{0} * 2^{n^2-n}$$

$$= 1 * 2^{n^2-n}$$

$$= \boxed{2^{n^2-n}}$$



Topic : Symmetric Relation

A relation R on set A is called a symmetric relation if and only if

if $a \, R \, b$ then $b \, R \, a$, $\forall a, b \in A$

i.e. if $(a, b) \in R$ then $(b, a) \in R$, $\forall a, b \in A$

for all pair of elements of set A, "if - then" statement must be true

Slide

eg: Let, $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ Empty relation is the smallest symmetric rel^n on any set A.

② $R_2 = \{(1,1)\}$ R_2 is symmetric { Presence or absence of diagonal order pair will never create any problem w.r.t. symmetricity }

③ $R_3 = \{(1,2), (2,1), (2,2)\}$ R_3 is symmetric

④ $R_4 = \{(1,1), (2,3), (3,1), (3,2)\}$

$(3,1) \in R_4$, but $(1,3) \notin R_4$
$\therefore R_4$ is not a symmetric Rel^n

- Empty relation on any set A is the smallest symmetric relation
- 'A×A' is the largest symmetric relation on any set A.

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "symmetric relation" are possible on set A where $|A| = n$.

Number of symmetric relation = (Any number of diagonal order Pairs are allowed) and (Possible choices w.r.t. non-diagonal Order Pairs)

- in non-diagonal order pair (a, b) we know $a \neq b$
- i.e. for non-diagonal order pair we need two distinct elements $a \& b$
- W.r.t two distinct elements $a \& b$ what are the different combinations of order pairs that may be present in a relation

Case (i)
both (a,b) & (b,a) are present in the relation

Case (ii)
(a,b) is present but (b,a) is not present in relation

Case (iii)
(a,b) is not present but (b,a) is present in relation

Case (iv)
Both (a,b) & (b,a) are not present in relation

Approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair (a,b), we know that $a \neq b$

  if $a=b$, then (a,b) is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

Case (i)	Case (ii)	Case (iii)	Case (iv)
both (a,b), (b,a) are present	(a,b) is present but (b,a) is not present	(a,b) is not present but (b,a) is present	Both (a,b) and (b,a) are not present

These are the only possible combinations of order pairs in a relation w.r.t. two distinct elements a & b

Approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair (a,b), we know that $a \neq b$

  if $a=b$, then (a,b) is a diagonal order pair.

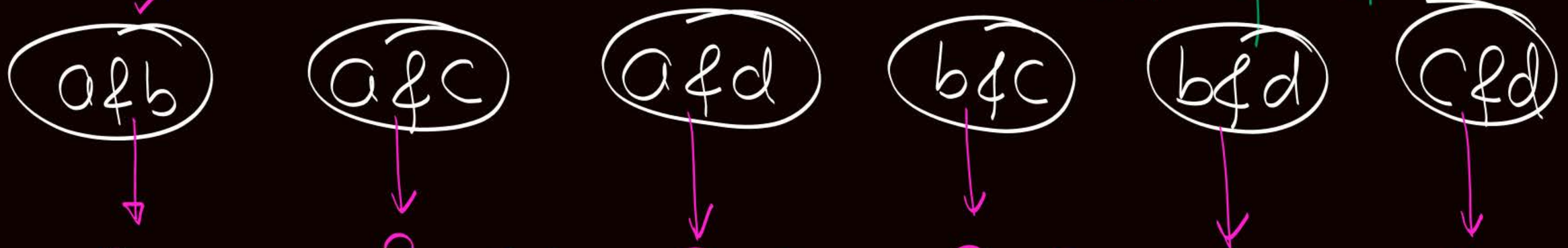
- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

Case (i)	Case (ii)	Case (iii)	Case (iv)
both (a,b), (b,a) are present	(a,b) is present but (b,a) is not present	(a,b) is not present but (b,a) is present	Both (a,b) and (b,a) are not present
Allowed in Symmetric relⁿ	Not Allowed in Symmetric relⁿ	Not Allowed in Symmetric relⁿ	Allowed in Symmetric relⁿ

Let $A = \{a, b, c, d\}$

$^4C_2 = 6$ ways to choose two distinct elements out of '4'.



2 and 2 and 2 and 2 and 2 and 2

$\therefore 2 * 2 * 2 * 2 * 2 * 2 = 2^6$

$= 2^{^4C_2}$

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "symmetric relation" are possible on set A where $|A| = n$.

Number of symmetric relation = (Any number of diagonal order Pairs are allowed) and (Possible choices w.r.t. non-diagonal Order Pairs)

$= ({}^nC_0 + {}^nC_1 + \ldots + {}^nC_n) * 2^{{}^nC_2}$

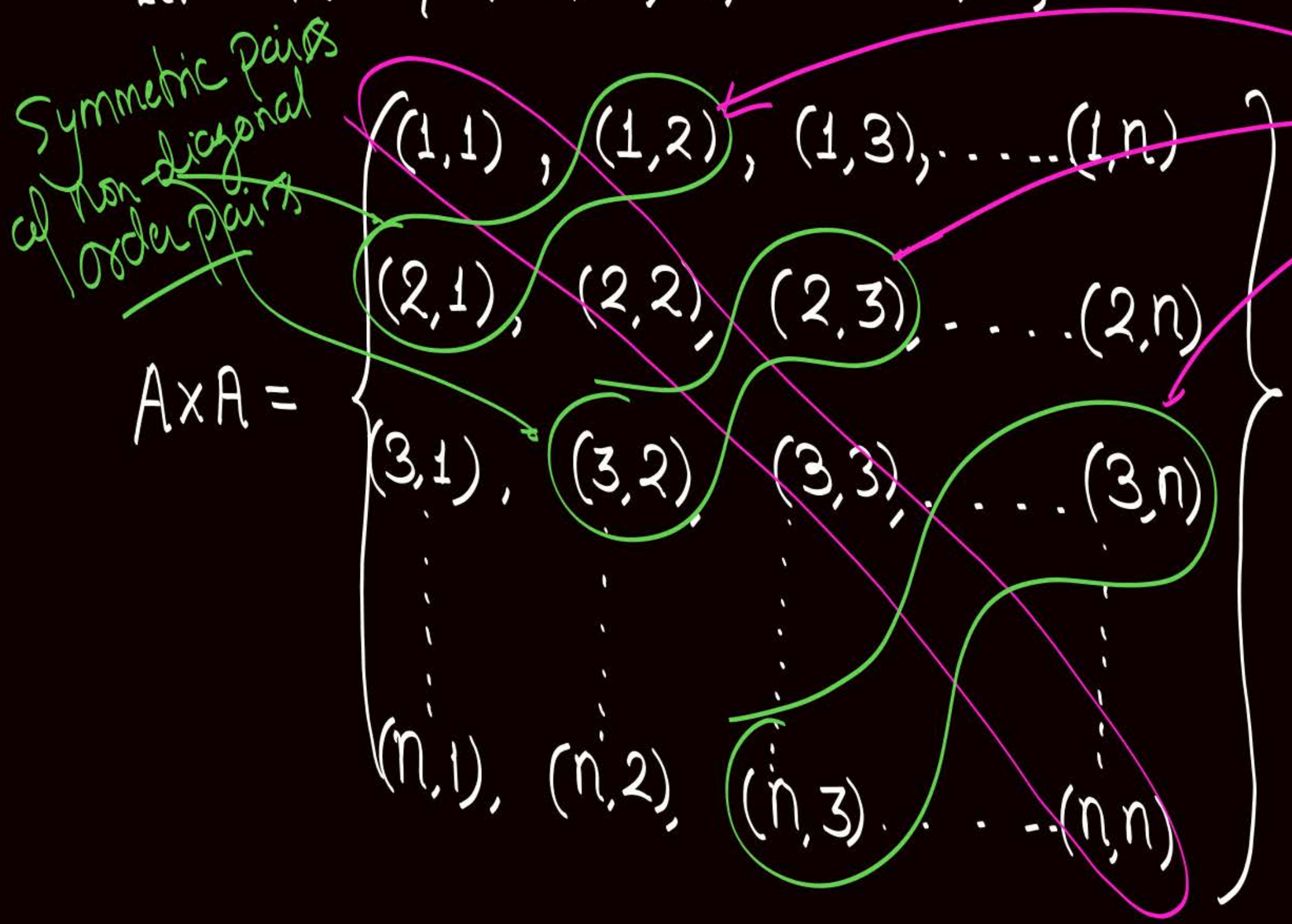
$= 2^n \times 2^{\frac{n^2-n}{2}}$

$= 2^{\frac{n^2+n}{2}} = 2^{\frac{n(n+1)}{2}}$

${}^nC_2$ ← We can Choose two distinct elements w.r.t. non-diagonal order pairs in ${}^nC_2$ ways

and for each pair of distinct elements we have '2' choices in a symmetric rel^n

Let $A = \{1, 2, 3, 4, \ldots, n\}$



No. of Symmetric Pairs of non-diagonal order pairs $= \frac{n^2 - n}{2}$

Symmetric Relⁿ = w.r.t diagonal & w.r.t non-diagonal

$= 2^n * 2^{\left(\frac{n^2-n}{2}\right)}$

$= 2^{\frac{n(n+1)}{2}}$



Topic : Anti-Symmetric Relation

A relation R on set A is called an anti-symmetric relation if and only if

$$\text{if } (a\,R\,b \text{ and } b\,R\,a) \text{ then } a=b, \quad \forall a,b \in A$$

i.e. $$\text{if } ((a,b)\in R \text{ and } (b,a)\in R) \text{ then } a=b, \quad \forall a,b \in A$$

If $(a=b)$ then "then" and "if-then" statement will be true and in that case "if-then" will always return true (i.e. we don't need to worry about if)

if $a=b$, then we are talking about diagonal order pairs.

i.e. w.r.t. diagonal order pairs if-then statement will always return true

i.e. No problem because of presence of diagonal order pairs

Slide

Note :-

In an anti-symmetric relation there will be no problem because of Presence or absence of any diagonal order pair.

But if $a \neq b$, i.e. when "then" statement is false then for "if-then" statement to be true our "if" statement itself must be false.

↳ i.e. In an anti-symmetric relation if $a \neq b$, then Either (i) $(a,b) \notin R$
or (ii) $(b,a) \notin R$
or (iii) $(a,b) \notin R$ and $(b,a) \notin R$

eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ Empty relation is the smallest anti-symmetric rel^n on any set A

② $R_2 = \{(1,1), (3,3)\}$ it is an anti-symmetric relation
it is a symmetric relation as well.

③ $R_3 : \{(1,1), (1,2), (3,2)\}$ $(2,3) \notin R_3$ ∴ okay
$(2,1) \notin R_3$ ∴ okay
R_3 is an anti-symmetric Rel^n

④ $R_4 = \{(1,2), (2,3), (3,1), (3,2)\}$

$(2,3) \in R_4$ & $(3,2) \in R_4$ but $2 \neq 3$ ∴ Not an anti-symm. rel^n!

if is true but then is false

Note: If a relation does not contain any non-diagonal order pair, then every such relation is symmetric as well as anti-symmetric

* Largest symmetric as well as anti-symmetric relation on set A is Δ_A

(diagonal relation)

Q:. Let $A = \{1, 2, 3, \ldots, n\}$
How many "anti-symmetric relation" are possible on set A where $|A| = n$.

Number of Anti-symmetric relation = (Any number of diagonal order Pairs may be Present) and (Possible Combinations w.r.t. non-diagonal order Pairs)

Approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair (a,b), we know that $a \neq b$

  if $a=b$, then (a,b) is a diagonal order pair.

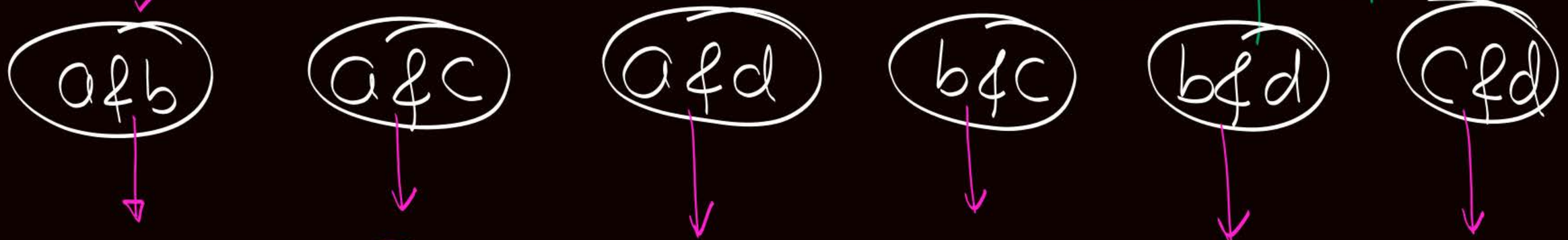
- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

Case (i)	Case (ii)	Case (iii)	Case (iv)
both (a,b), (b,a) are present	(a,b) is present but (b,a) is not present	(a,b) is not present but (b,a) is present	Both (a,b) and (b,a) are not present
Not Allowed in Anti-Symmetric rel^n	Allowed in anti-Symmetric rel^n	Allowed in anti-Symmetric rel^n	Allowed in anti-Symmetric rel^n

Let $A = \{a, b, c, d\}$

$^4C_2 = 6$ ways to choose two distinct elements out of '4'.



3 choices $\times$ 3 $\times$ 3 $\times$ 3 $\times$ 3 $\times$ 3 $= 3^6$

$= 3^{n_{C_2}}$

{(a,b) ~~(b,a)~~}

{~~(a,b)~~ (b,a)}

{~~(a,b)~~ ~~(b,a)~~}

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "anti-symmetric relation" are possible on set A where $|A| = n$.

Number of Anti-symmetric relation = (Any number of diagonal order Pairs may be Present) and (Possible Combinations w.r.t. non-diagonal order Pairs)

$= 2^n * 3^{(n_{C_2})}$

we can choose '2' distinct elements out of 'n' in n_{C_2} ways

For each Pair of distinct elements '3' choices of order pairs in an anti-symmetric rel^n.

$= 2^n \cdot 3^{\frac{n^2-n}{2}}$



Topic : Asymmetric Relation

A relation R on set A is called asymmetric relation if and only if

if $a \overset{R}{b}$ then $b \overset{\not R}{a}$, $\forall a, b \in A$

i.e. if $(a,b) \in R$ then $(b,a) \notin R$, $\forall a, b \in A$

Does not matter wheather 'a=b' or not

∴ in asymmetric relation diagonal order pairs are not allowed.

Slide

eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$ Empty relation is the smallest Asymmetric rel^n on any set A

② $R_2 = \{(1,2), (3,3)\}$

✗ diagonal not allowed ∴ Not an asymmetric Rel^n

③ $R_3 = \{(1,2), (2,3), (3,1), (3,2)\}$

∴ Not an asymmetric rel^n

④ $R_4 = \{(1,2), (3,1), (3,2)\}$ It is an asymmetric Rel^n

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "asymmetric relation" are possible on set A where $|A| = n$.

Number of asymmetric relation = (None of the diagonal order Pair is allowed) and (Possible Combinations w.r.t. non-diagonal order Pairs)

Approach w.r.t. non-diagonal order pairs

- In a non-diagonal order pair (a,b), we know that $a \neq b$

  if $a=b$, then (a,b) is a diagonal order pair.

- For a non-diagonal order pair we will require two distinct elements 'a' and 'b' s.t. $a \neq b$

- W.r.t. two distinct elements 'a' and 'b' possible combinations of order pairs in an arbitrary relation

Case (i)	Case (ii)	Case (iii)	Case (iv)
both (a,b), (b,a) are present	(a,b) is present but (b,a) is not present	(a,b) is not present but (b,a) is present	Both (a,b) and (b,a) are not present
Not Allowed in Asymmetric rel^n	Allowed in Asymmetric rel^n	Allowed in Asymmetric rel^n	Allowed in Asymmetric rel^n

Q:. Let $A = \{1, 2, 3, \ldots, n\}$

How many "asymmetric relation" are possible on set A where $|A| = n$.

Number of asymmetric relation = (None of the diagonal order Pair is allowed) and (Possible Combinations w.r.t. non-diagonal order Pairs)

$= {}^{n}C_0 * 3^{({}^{n}C_2)}$ } ← Same as anti-symmetric

$= 1 * 3^{\frac{n(n-1)}{2}}$

$= \boxed{3^{\left(\frac{n^2-n}{2}\right)}}$

$\{(a,b), (b,a)\}$

$\{(a,b)\}$

Note: ① On a set A, Empty relation is the only relation which is symmetric, anti-symmetric as well as asymmetric

② Any subset of an anti-symmetric relation is also anti-symmetric

③ Every asymmetric relation is anti-symmetric as well, but Every anti-symmetric relation need not be an asymmetric relation

Note: (4) Any subset of asymmetric relation is also an asymmetric relation



Topic : Transitive Relation

A relation R on set A is called transitive relation if and only if

if $(a R b$ and $b R c)$ then $a R c$, $\forall a, b, c \in A$

i.e. if $((a,b) \in R$ and $(b,c) \in R)$ then $(a,c) \in R$, $\forall a, b, c \in A$

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H.W

eg:- Let $A = \{1, 2, 3\}$

① $R_1 = \{\ \}$

② $R_2 = A \times A$

③ $R_3 = \{(1,1), (2,2)\}$

} check for transitive relation

H.W

let $A = \{1, 2, 3\}$

④ $R_4 = \{(1,2), (2,1), (1,1)\}$ Check whether R_4 is transitive or not

H.W

Eg. $A = \{1, 2, 3\}$

$R_5 = \{(1,2), (1,3)\}$ transitive or not ?

H.W.

eg: $A = \{1, 2, 3\}$

$R_6 = \{(2,3), (1,2)\}$ transitive or not ?

H.W

eg 1: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

transitive or not ?

H.W.

eg 2: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

transitive or not ?

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$

transitive or not?

eg. Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

transitive or not?



2 mins Summary

Topic Reflexive relation

Topic Irreflexive relation

Topic Symmetric relation

Topic Anti-symmetric relation

Topic Asymmetric relation

Topic Transitive relation

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THANK - YOU

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